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VERSION 0.0

Risk Assessment

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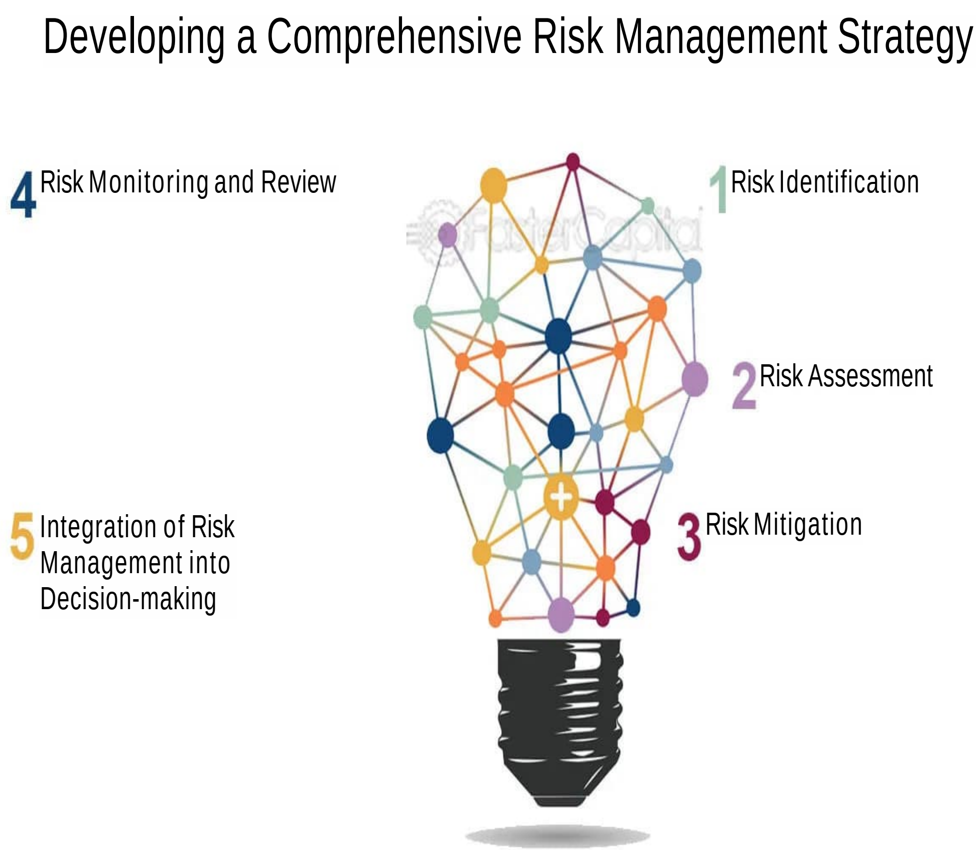
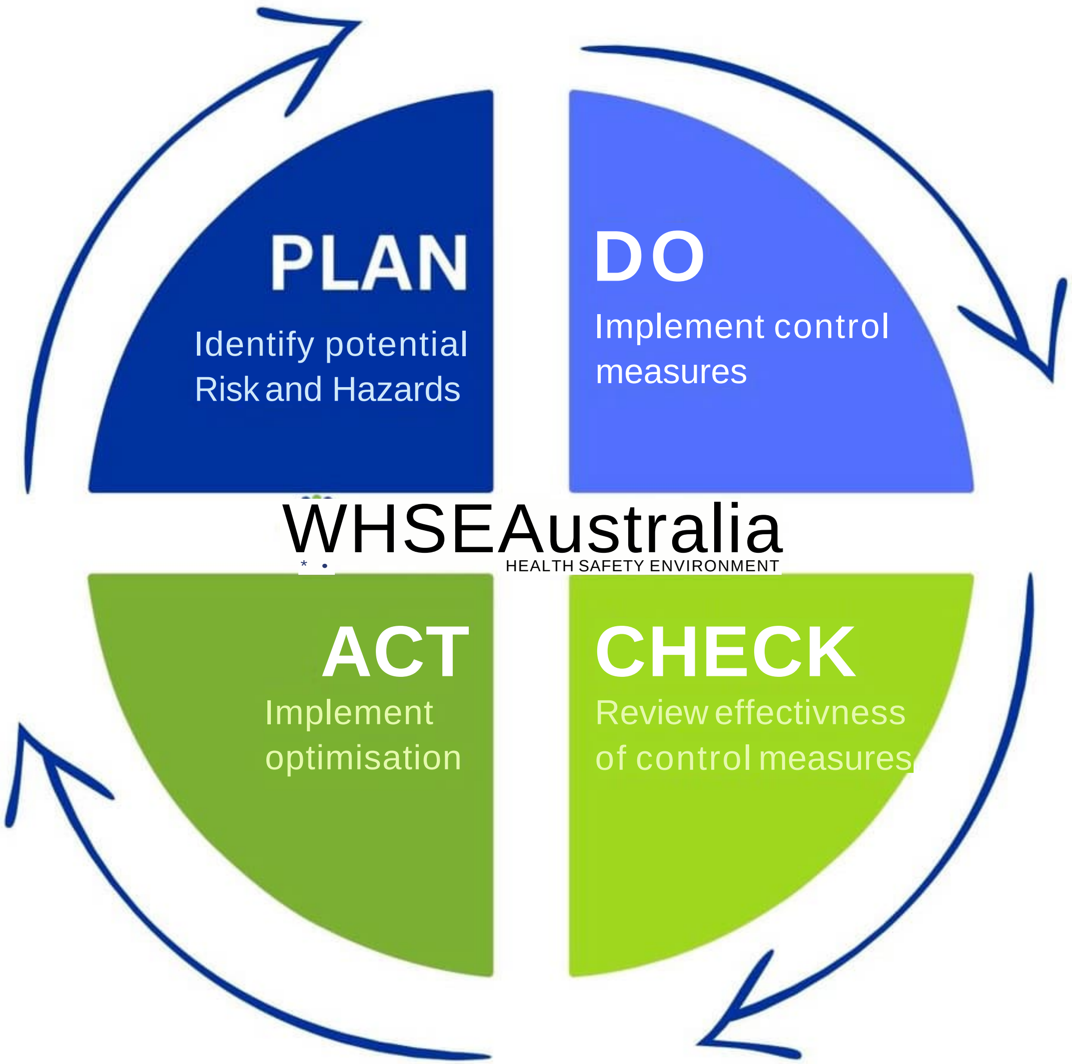
عملية تقييم المخاطر

1. Hazard Identification
2. Risk Assessment (Likelihood and Severity)
3. Implementation of Control Measures
4. Monitoring and Review
5. Continuous Improvement



PDCA Cycle

PDCA in Risk Assessment



Hazard Identification Techniques

1 Brainstorming Sessions

Gathering diverse perspectives from cybersecurity specialists, AI researchers, and HSE professionals facilitates comprehensive hazard identification. Encourage open discussions and creative thinking to uncover potential risks that may be overlooked otherwise.

2 Checklist Analysis

Leverage industry-standard checklists, customized for cybersecurity and AI, to systematically assess potential hazards. Ensure checklists are regularly updated to reflect evolving threats and emerging technologies. Prioritize items based on likelihood and potential impact.

3 Incident Investigation

Thoroughly investigate past cybersecurity incidents and AI-related failures to identify underlying hazards and contributing factors. Implement corrective actions to prevent recurrence and improve overall safety and security protocols. Share lessons learned across the organization.

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Job Hazard analysis

Probability (P) احتمالية الحدوث

Oe*Lail ,L, —i =-ii	Likel iho od الاحتمالات	Weight وزن نسبي
عملية دائمة Continuous process	متكرر Fnequent	s
عملية يومية Daily process	مرجح Likely	q.
a_ أسبوعية a_i_.xjz Weekly process	متباعد Occasion 3 1	3
.يهر1 j-.i jtST / 3 .- .-:- عملية r lonttily process / rcnonttis	نادرا Sei do m	2
أكثر من سنة M ore th3n -a year	غير محتمل Unlikely	1

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Job Hazard analysis

Severity (Sj UijAR

Assets sZjK-7fl1'!	Environment بيئة	Health الصحة	Safety lajLJI	level المستوى	Weight 41* نسبي
JIkJI بالشركة jl bi Extens ve damage to comcany	j ikjpo jjijJ 1. '' fL تأثير Ub Impact penmanent/longer lasting damage off site	مرض ـــ Occupational disease	وفاة شخص أو أكثر Fatality, single or multiole	كارثية Catastrophic	5
أضرار بالقسم Damage to Section	تحتاج aL*.~jill Tj-S 2_1 Jj*2> -J j jdb لتحقيق وإجراءات تصحيحية impact nmnea carnage on sire leading toinvestigations & corrective =ir+ir»•"r	مرض دائم Permanent disease	عجز جزئي / كامل Partial / Comp ete incaparity	كبيرة Major	4
أضرار بالمعدات Damage to ecuipment	T--= - ضرر محدود J* لإجراءات تصحيحية Impact limited damage on site leading to corrective acrons	JjjJ مرض يسبب حالة إيقاف أيام عن (LWC) Lost Workday Case to disease	JixJI pu: -jiij J .ITfl 3J1=- ‘LWC) Lost Workday Case	حرجة Critical	3 3
أضرار بالبنية التحتية Damage to infrastnualire	لحظياً L-Jc ■<,< -ll pSSliUI JiJj Ji± Impact on site release immediately contained	مرض يسبب حالة علاج طبي أو تغيير نوع J i (MTC) Or (FLWC) to occupational disease	حالة علاج طبي أو تغيير نوع العمل (MTC] Medical Treatment Case, Or (RWCj Restricted Workday Case	محدودة Moderate	2 2
بالاثاث j i Damage of furniture	jut 1JH-4 J-luL Ij . Jc_—i جانب بيئي Aspect environmental very simple with no impact	مرض jjlr Transient disease	iJL_iq حادث / 3_.; إسعافات A.ls- First Aid / Near miss case	ضئيلة Negligible	1 1

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Job Hazard analysis

Guide to Rankings دليل التصنيفات

هام Significant	إيقاف التشغيل ومراجعة ضوابط التحكم Stop operation & review controls.	مقبولة _.-X jjz-'Si-a Unacceptable tisks	17 - 25
	iZjljjl r ■—وجود مر '-<• Li14 استمرار العمل بحذر ■! jSCuj T.،n~' الإجراءات 2LAS'- j uZ.- js-yL' ،pbl 7 -j7■■"■* i '■*'-j~ الفور يتم ~ .j:'-1 IjLusI ضوابط .'- ~ العمل، وت High priority remedial action, Proceed with extreme caution with pi present at all times. Implement additional |secardary| controls immediately. Review within 7 days.	مخاطر عالية High risk	11-16
	30 pj ،1 ' . 9 4_ LZI ضوابط .i ■-:~j jfl |_uut_'-jlj a~ir ■:■■rj|~all 9 3 -*- i - .jiT'l iZ:<.l - Jl J jl يوما. Take remedial action at appropriate time, Proceed with care, add'tiona control is advised. Review shall be implemented withiin 30 days.	مخاطر محدودة Limited risk	4-10
غير هام Non-5 gniftan:	المواد، J,.MZjlcJ q-yi Ji i ■■■ -• t-1 -IjJZ —i— C J|~|~■■■' |3JX 0-1 1 المخاطر -LfOJ ■—ي أو البيئة. Risk reduction should be further considered, particularly severity. Especially changes in procedures, materials or environment	مخاطر -cr.Tz-3 ■j-'-1 Risk can be accepted	1-3

ملحوظة: مستويات الأهمية j Note:Significant levels

Mi 2—'■c'jr-./ أو خرق 4 z M-j- j 3 :1■x.-yhJi jyC j Environmenta issues shall be any store 4 or breach of legislative requirements

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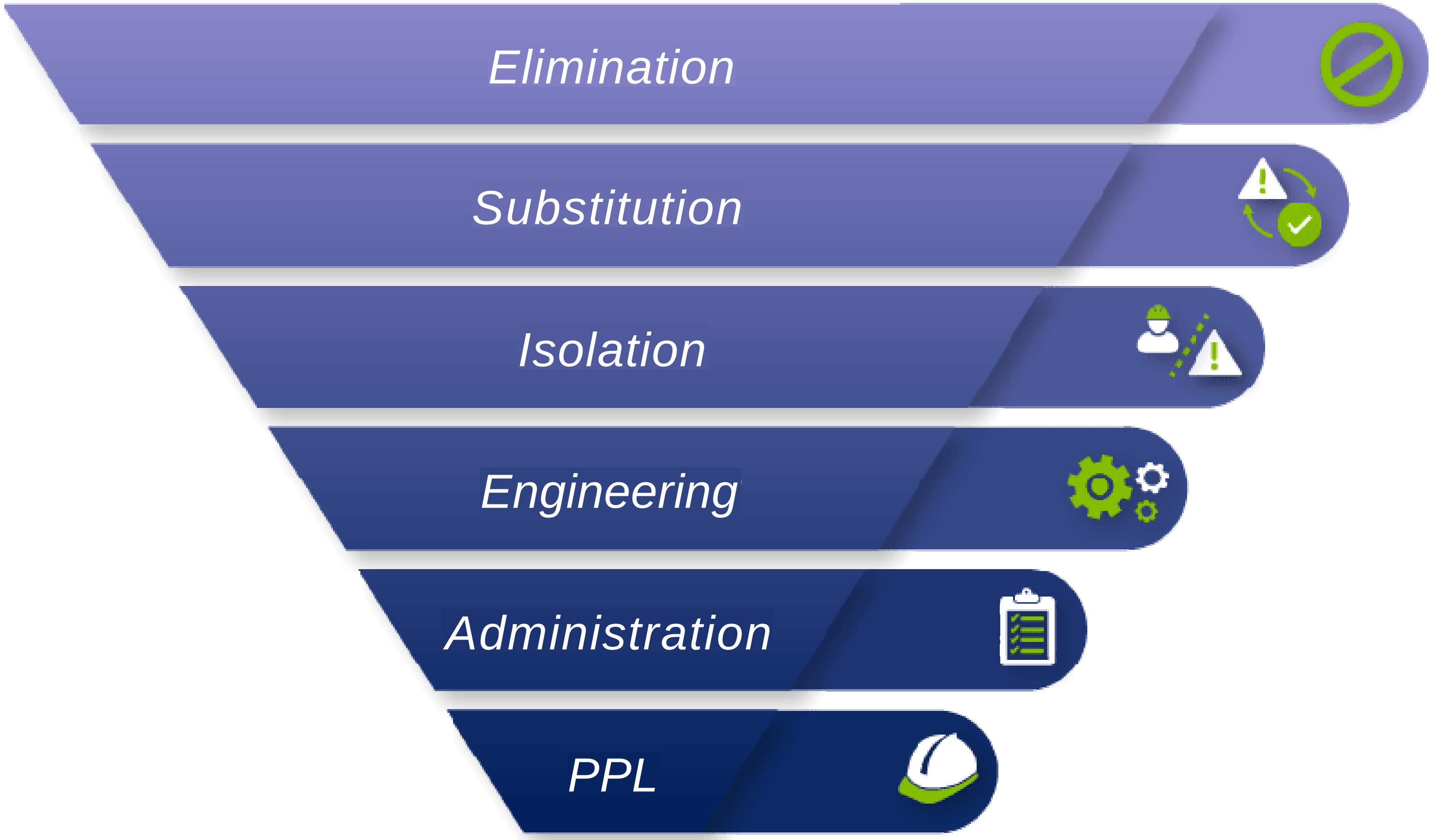
مصفوفة المخاطر

5x5 RISK MATRIX

SEVERITY

LIKELIHOOD	1	2	3	4	5
1	LOW 1	LOW 2	LOW 3	MEDIUM 4	MEDIUM 5
2	LOW 2	MEDIUM 4	MEDIUM 6	HIGH 8	HIGH 10
3	LOW 3	MEDIUM 6	HIGH 9	HIGH 12	EXTREME 1 5
4	MEDIUM 4	HIGH 8	HIGH 12	HIGH 16	EXTREME 20
5	MEDIUM 5	HIGH 10	EXTREME 15	EXTREME 20	E XTR EME 25

Herachicy of control



Smart control measures

SMART

SPECIFIC

Be clear and specific so your goals are easier to achieve. This also helps you know how and where to get started!

MEASURABLE

Measurable goals can be tracked, allowing you to see your progress. They also tell you when a goal is complete.

ACTIONABLE

Are you able to take action to achieve the goal? Actionable goals ensure the steps to get there are within your control.

REALISTIC

Avoid overwhelm and unnecessary stress and frustration by making the goal realistic.

TIMEBOUND

A date helps us stay focused and motivated, inspiring us and providing something to work towards.

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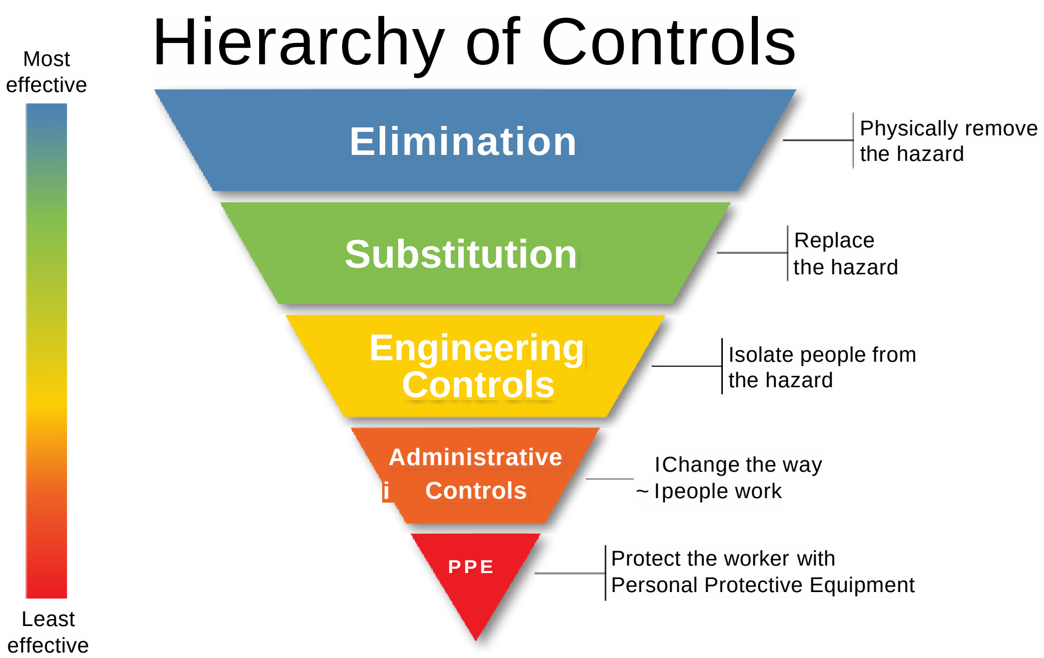
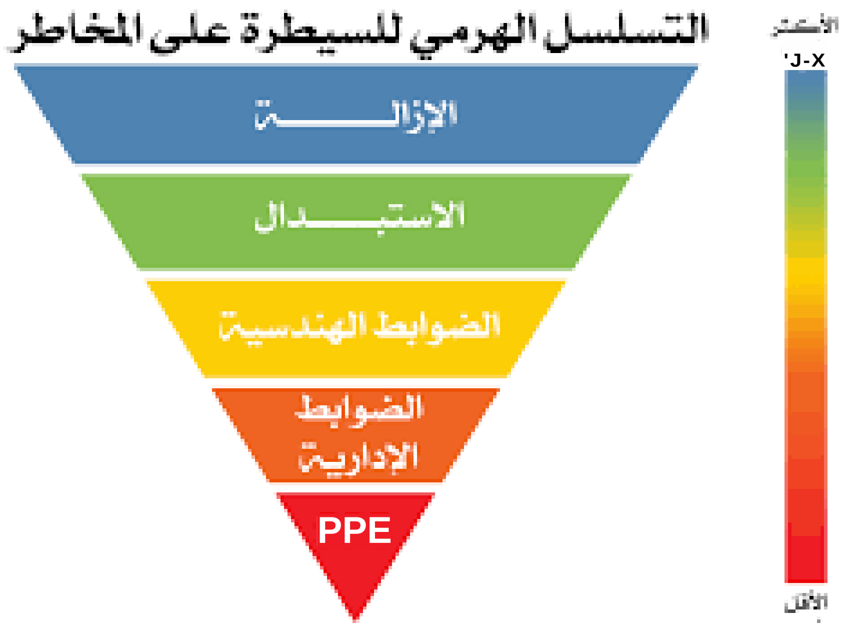
Job Hazard analysis

وسائل التحكم في العواقب.





Control Measures: A Hierarchy of Effectiveness

1

Elimination

Remove the hazard completely. Examples: Discontinue use of vulnerable software or hardware, eliminate unnecessary data collection, or replace a risky AI algorithm with a simpler, safer alternative. This is the most effective control.

2

Substitution

Replace the hazard with a less dangerous alternative. Examples: Use a more secure programming language, substitute a less complex AI model, or migrate to a more robust cloud platform with enhanced security features.

3

Engineering Controls

Implement physical or technical controls to isolate the hazard. Examples: Install firewalls, intrusion detection systems, implement strong access control mechanisms, or deploy AI-powered security solutions to detect and prevent cyberattacks.

4

Administrative Controls

Establish procedures and policies to minimize exposure to the hazard. Examples: Implement security awareness training programs, conduct regular vulnerability assessments, enforce strong password policies, or establish incident response plans.

5

Personal Protective Equipment (PPE)

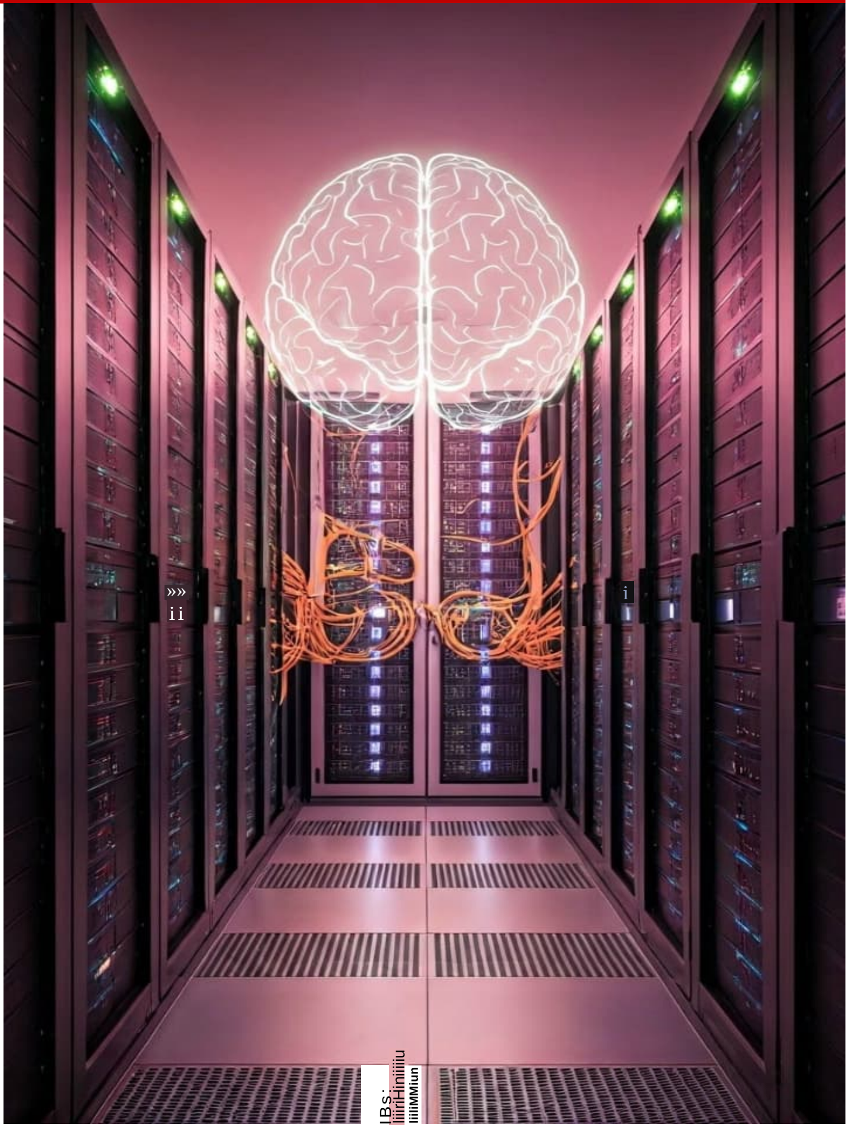
Provide employees with equipment to protect them from hazards. Examples: Use ergonomic workstations, provide anti-glare screen filters, or mandate the use of secure VPN connections when accessing sensitive data remotely.

The relationship between risk assessment, cybersecurity, and AI

Risk identification, assessment, and control:

A cybersecurity and artificial intelligence perspective.

- This lecture covers risk identification, assessment, and control, with a particular focus on the dynamic fields of cybersecurity and artificial intelligence.
- We will explore how traditional occupational safety and health principles apply and adapt to the unique challenges posed by these technologies.
- Our goal is to equip you with the knowledge and tools needed to proactively manage risks, ensuring a safer and more secure digital future.



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Risk management

Risk Identification in the Cybersecurity and AI Scope

❖ Traditional Risks

- **Physical risks** remain relevant.
- An uncomfortable work environment can lead to musculoskeletal disorders.
- **Electrical hazards** from damaged equipment pose safety threats.
- **Fire** risks in server rooms require strong preventive measures.
- These well-known risks demand continuous attention in our digital environment

❖ Cybersecurity-Specific Risks

- **Data breaches** pose a significant risk, leading to financial losses and reputational damage.
- **Malware infections** can disrupt operations and expose sensitive information to danger.
- **Social engineering** attacks exploit human vulnerabilities, highlighting the importance of security awareness training and phishing simulation exercises.

Monitoring and Reviewing Control Effectiveness

1

Key Performance Indicators (KPIs)

Establish metrics to track the effectiveness of implemented control measures. Examples: Number of detected cyberattacks, time to patch vulnerabilities, user compliance with security policies, or AI model accuracy and fairness.

2

Regular Audits and Inspections

Conduct periodic audits and inspections to assess the implementation and effectiveness of control measures. Identify areas for improvement and implement corrective actions to address deficiencies.

3

Feedback Mechanisms

Establish channels for employees to report potential hazards, security incidents, or concerns related to AI systems. Encourage open communication and foster a culture of safety and security awareness. Anonymous reporting is a must.

Home Work:

Risk Assessment

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